

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Project Documentation

Team Detail	Information
Team ID	SWTID-2026-3648
Team Size	5
Team Leader	S SANTHOSH
Team Member	Durainesan
Team Member	Sujith.R
Team Member	Sivaprakash
Team Member	Gokulavanan A
Platform	ServiceNow (Flow Designer, Service Catalog)

1. Brainstorming & Ideation Phase

1.1 Problem Statement

In many organizations, employees request standard laptops through emails, chats or phone calls. Requests get lost, managers approve informally, procurement teams do not see a clear queue, and requesters have no visibility into the status of their order. This manual process is slow, error-prone and hard to audit.

The project aims to automate the standard laptop ordering process in ServiceNow using Flow Designer, so that every request is submitted through one catalog form, routed for approval automatically, fulfilled through tracked tasks, and communicated back to the requester without manual follow-up.

1.2 Empathy Map

Quadrant	Employee (Requester)	IT Procurement Agent
Says	"When will my laptop arrive?" "Who approved my request?"	"Which requests are approved?" "What model was ordered?"
Thinks	Wants a simple, single place to raise the request.	Wants complete details in one record without chasing emails.
Does	Sends emails, follows up repeatedly with manager and IT.	Searches inboxes, creates orders manually, updates spreadsheets.
Feels	Frustrated by delays and lack of visibility.	Overloaded and worried about missing or duplicate requests.

1.3 Brainstorming

The team discussed the following ideas for improving the laptop request process:

- A Service Catalog item with a fixed list of standard laptop models.
- Automatic manager approval before procurement starts.
- Automatic creation of a fulfillment task for the procurement team.
- Email notifications to the requester on approval, rejection and completion.
- A dashboard to report request volume and turnaround time (future scope).

1.4 Idea Prioritization

Idea	Impact	Effort	Decision
Catalog item for standard laptops	High	Low	Implement
Flow Designer approval and task automation	High	Medium	Implement
Requester notifications	Medium	Low	Implement
Reporting dashboard	Medium	High	Future scope

2. Requirement Analysis Phase

2.1 Functional Requirements

ID	Requirement	Description
FR-1	Laptop request catalog item	Employees can order a standard laptop from the Service Catalog.
FR-2	Request variables	Form captures laptop model, business justification, needed-by date and requested-for user.
FR-3	Automatic trigger	Flow starts automatically when a catalog request is submitted.
FR-4	Manager approval	Request is sent to the requester's manager for approval.
FR-5	Fulfillment task	On approval, a catalog task is created and assigned to the procurement group.
FR-6	Rejection handling	On rejection, the request is closed and the requester is notified.
FR-7	Notifications	Requester is informed at approval, rejection and completion stages.

2.2 Non-Functional Requirements

- Usability: the form must be simple and completed in under two minutes.
- Reliability: every submitted request must trigger the flow exactly once.
- Traceability: every approval and task must be visible on the request record.
- Maintainability: the flow must be built with low-code Flow Designer actions so it can be changed without scripting.

2.3 Users and Roles

Role	Responsibility
Requester (Employee)	Submits the laptop request and tracks its status.
Manager	Approves or rejects the request.
Procurement Agent	Works the fulfillment task and orders the laptop.
Administrator	Builds and maintains the catalog item and flow.

2.4 Technology Stack

Component	Purpose
ServiceNow Instance (PDI)	Development platform.
Service Catalog	Front end where the request is raised.
Flow Designer	Low-code automation of approval, tasks and notifications.
Approval Engine	Handles manager approval records.

Component	Purpose
Notifications	Email updates to the requester.

3. Project Design Phase

3.1 Solution Architecture

The solution uses the standard ServiceNow request model: a catalog request (REQ) contains a requested item (RITM), which triggers the flow. The flow drives approval, task creation and notifications, and updates the RITM state at each stage.

3.2 Process Flow

1. Employee opens the Service Catalog and selects "Standard Laptop Request".
2. Employee fills in the variables and submits; a REQ and RITM are created.
3. Flow Designer trigger (Service Catalog) starts the flow.
4. Flow asks for approval from the requester's manager.
5. If approved: a catalog task is created for the procurement group and the requester is notified.
6. If rejected: the RITM is closed as rejected and the requester is notified.
7. When the procurement task is closed, the RITM is marked Closed Complete and the requester receives a completion notification.

3.3 Catalog Item Design

Variable	Type	Details
Requested For	Reference (User)	Defaults to the logged-in user.
Laptop Model	Select Box	Standard approved models (for example Dell Latitude, HP EliteBook, Lenovo ThinkPad).
Business Justification	Multi Line Text	Mandatory; reason for the request.
Needed By	Date	Expected delivery date.
Additional Notes	Single Line Text	Optional.

3.4 Flow Design

Step	Element	Configuration
1	Trigger: Service Catalog	Catalog item = Standard Laptop Request; runs when the item is ordered.
2	Action: Get Catalog Variables	Retrieves the laptop model, justification and needed-by date.
3	Action: Ask for Approval	Approver = requester's manager; rule = approved when manager approves.
4	Flow Logic: If	Branch on approval state (Approved / Rejected).
5a	Action: Create Catalog Task	Assignment group = Procurement; short description includes the laptop model.
5b	Action: Update Record	On rejection, set RITM state to Closed Incomplete.
6	Action: Send Email	Notify the requester of the outcome.
7	Action: Update Record	When the task closes, set RITM to Closed Complete.

3.5 User Stories

ID	User Story	Priority
US-1	As an employee, I want to order a standard laptop from one form so that I do not have to send emails.	High
US-2	As a manager, I want to receive an approval request automatically so that I can respond quickly.	High
US-3	As a procurement agent, I want a task with all order details so that I can place the order without follow-up.	High
US-4	As an employee, I want email updates so that I know the status of my request.	Medium

4. Project Planning Phase

4.1 Approach

The project is planned in short sprints, with each sprint delivering a testable part of the solution. The team leader coordinates the work, and all members take part in design, configuration and testing.

4.2 Sprint Plan

Sprint	Goals	Deliverable	Duration
Sprint 1	Ideation, requirements, instance setup	Problem statement, requirement list, PDI ready	2 days
Sprint 2	Design and catalog item creation	Solution design, catalog item with variables	2 days
Sprint 3	Flow Designer build	Working flow with approval, task and notifications	3 days
Sprint 4	Testing and fixes	Test results and corrected flow	2 days
Sprint 5	Documentation and demonstration	Final report and demo	2 days

Durations are indicative and can be adjusted to the team's schedule.

4.3 Risks and Mitigation

Risk	Mitigation
Manager not set on the requester's user record	Configure the manager field on test users before testing.
Flow not triggering	Confirm the flow is activated and the catalog item name matches the trigger.
Notification not delivered in PDI	Check the email log and notification activation.

5. Project Development Phase

This phase describes the configuration performed in the ServiceNow instance. Screenshots of each step can be inserted under the corresponding heading.

Step 1: Prepare the Instance and Users

- Log in to the ServiceNow developer instance as an administrator.
- Create or verify a test requester and a manager user; set the manager on the requester's record.
- Create a group named Procurement and add a member.

Step 2: Create the Catalog Item

- Navigate to Service Catalog > Catalog Definitions > Maintain Items and click New.
- Name the item "Standard Laptop Request" and add a short description and category.
- Add the variables listed in section 3.3 and mark Business Justification as mandatory.
- Set the item to Active and publish it to the catalog.

Step 3: Build the Flow in Flow Designer

- Open Flow Designer and create a new Flow named "Standard Laptop Order Automation".
- Add the Service Catalog trigger and select the Standard Laptop Request item.
- Add the Get Catalog Variables action and select the variables to use.

Step 4: Add Approval Logic

- Add the Ask for Approval action on the requested item record.
- Set the approver to the requester's manager and the rule to approved when the manager approves.
- Add an If condition on the approval state to branch into Approved and Rejected paths.

Step 5: Fulfillment and Notifications

- Approved path: add Create Catalog Task, assign it to the Procurement group, and include the laptop model and justification.
- Rejected path: add Update Record to close the RITM as incomplete.
- Add Send Email actions to notify the requester at each outcome.
- Add a final step to set the RITM to Closed Complete once the task is closed.

Step 6: Activate the Flow

- Save the flow, click Activate, and confirm its status shows Active.

6. Project Testing Phase

6.1 Test Approach

Testing was planned as functional end-to-end testing from the requester's point of view, supported by the Flow Designer execution details to confirm each action ran.

6.2 Test Cases

ID	Scenario	Steps	Expected Result	Status
TC-01	Submit laptop request	Order the catalog item with all required fields.	REQ and RITM created; flow starts.	
TC-02	Mandatory field check	Submit without business justification.	Form blocks submission.	
TC-03	Manager approval	Manager approves the request.	Catalog task created for Procurement.	
TC-04	Manager rejection	Manager rejects the request.	RITM closed incomplete; requester notified.	
TC-05	Task completion	Procurement closes the task.	RITM set to Closed Complete; completion email sent.	
TC-06	Flow execution log	Open Flow Designer execution details.	All actions show Completed.	

Update the Status column with Pass or Fail once each case has been executed, and attach screenshots as evidence.

7. Project Documentation Phase

7.1 Deliverables

Document	Content
Ideation document	Problem statement, empathy map, brainstorming outcomes.
Requirement document	Functional and non-functional requirements, roles, technology.
Design document	Architecture, process flow, catalog and flow design, user stories.
Plan	Sprint plan and risk register.
Development record	Configuration steps with screenshots.
Test report	Test cases and results.
Demonstration	Recorded or live demo of the working solution.

7.2 Benefits

- Single, consistent entry point for laptop requests.
- Faster approvals and fulfilment through automatic routing.
- Full audit trail of approvals and tasks.
- Improved requester experience through status notifications.

7.3 Future Scope

- Extend the pattern to other standard hardware such as monitors and mobiles.
- Add automatic asset record creation when the laptop is delivered.
- Build a dashboard for request volume and turnaround time.

8. Project Demonstration Phase

8.1 Demonstration Scenario

1. Log in as the requester and open the Service Catalog.
2. Order the Standard Laptop Request with sample values.
3. Show the generated REQ and RITM records.
4. Open Flow Designer > Executions and show the flow started.
5. Log in as the manager and approve the request.
6. Show the catalog task created for the Procurement group.
7. Close the task as the procurement agent and show the RITM as Closed Complete.
8. Show the notification emails sent to the requester.

8.2 Demonstration Checklist

Item	Ready
Flow is active	
Test users and manager relationship configured	
Procurement group has a member	
Screenshots captured for each phase	
Demo video link added	

Demo video / presentation link: _____

8.3 Conclusion

The project shows how Flow Designer can replace an email-driven laptop request process with a structured, automated workflow. By combining a Service Catalog item, manager approval, fulfilment tasks and notifications, the solution reduces manual effort, improves visibility for requesters, and gives the IT procurement team a reliable queue to work from.